

CYBERSECURITY FUNDAMENTALS

COMPLETE PROFESSIONAL VISUAL STUDY MANUAL

Defensive Security Edition - Paid Enrollee Course Material

Course Code: EDA-CYB-114

Level: Beginner to Intermediate

Duration: 6 Weeks (18 Lessons)

Certificate: Professional Certificate in Cybersecurity Fundamentals

Prerequisite: Computer Appreciation / Digital Literacy

Learning Format: Classroom + Safe Guided Practical + Case Studies

COURSE OVERVIEW

Cybersecurity Fundamentals introduces learners to the safe and responsible protection of devices, accounts, networks and information. The course focuses on defensive security, risk awareness, secure habits, incident reporting, data protection and professional cybersecurity pathways.

Practicals use safe demonstrations, fictional examples, account-security checks and approved training scenarios. Learners are not asked to attack real systems, bypass security controls or obtain other people's credentials.

LEARNING OBJECTIVES

- Explain confidentiality, integrity and availability.
- Identify assets, threats, vulnerabilities, risks and security controls.
- Protect accounts with strong unique passwords and multi-factor authentication.
- Recognise phishing and social-engineering warning signs.
- Understand common malware categories and defensive measures.
- Apply safe update, device and network practices.
- Protect personal and organisational information.
- Understand backups and recovery.
- Apply least privilege and role-based access concepts.
- Respond appropriately to suspected incidents.
- Understand privacy, ethics and responsible cybersecurity practice.
- Identify professional cybersecurity career pathways.

COURSE OUTLINE

Lesson	Topic
1	Cybersecurity Foundations
2	Assets, Threats, Vulnerabilities and Risk
3	Passwords, Passphrases and MFA
4	Phishing and Social Engineering Awareness
5	Malware and Endpoint Protection
6	Updates, Patching and Secure Configuration
7	Safe Internet, Browser and Network Use
8	Data Protection, Privacy and Classification
9	Backups and Recovery
10	Access Control and Least Privilege
11	Email and Messaging Security
12	Mobile Device and Cloud Security
13	Security Awareness for Organisations
14	Incident Recognition, Reporting and Response
15	Cybersecurity Policies, Ethics and Law
16	Risk Assessment and Business Continuity
17	Cybersecurity Careers and Defensive Tools
18	Final Defensive Security Project

LESSON 1: CYBERSECURITY FOUNDATIONS

Cybersecurity protects digital systems and information from unauthorized access, disruption, alteration and loss.

Cybersecurity Foundations

Confidentiality, Integrity and Availability

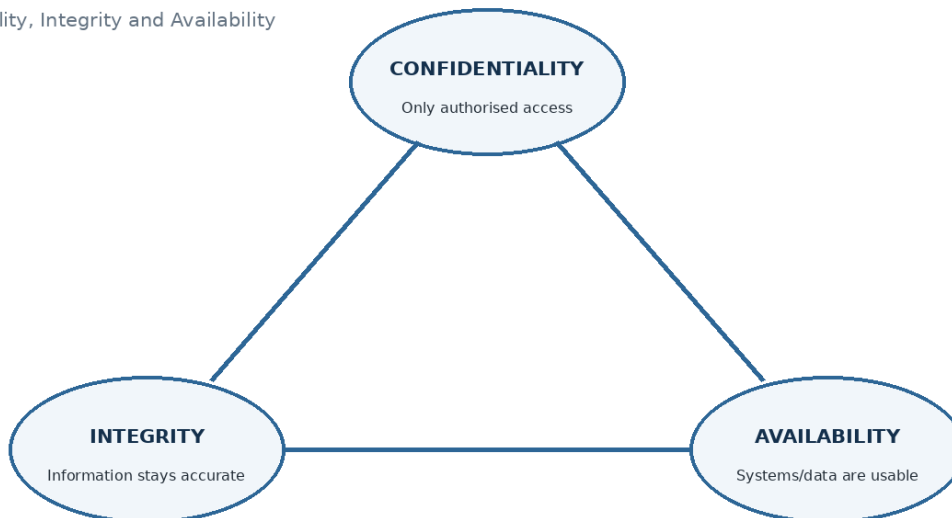


Figure 1. The confidentiality, integrity and availability security model.

The CIA Triad

- Confidentiality - information is available only to authorised people.
- Integrity - information remains accurate and is changed only in authorised ways.
- Availability - systems and information are accessible when authorised users need them.

People, Process and Technology

Security depends on responsible users, clear procedures and appropriate technical controls working together.

Everyday Cybersecurity

Examples include locking a device, using unique passwords, checking suspicious messages, updating software and backing up important work.

Safe Guided Practical

1. Identify five digital assets you use for school or work.
2. For each asset, state why confidentiality, integrity or availability matters.
3. Create a personal safe-computing checklist without including passwords.

Review Questions

1. What is cybersecurity?
2. Explain confidentiality.

3. Explain integrity.
4. Explain availability.

Assignment

Prepare a one-page summary of the CIA triad.

LESSON 2: ASSETS, THREATS, VULNERABILITIES AND RISK

Risk becomes easier to manage when we distinguish what we value, what can cause harm, where weaknesses exist and which controls reduce exposure.

Threat, Vulnerability and Risk

A simple defensive risk model



Figure 2. Assets, threats, vulnerabilities and controls.

Assets

Assets include devices, accounts, files, services, reputation and business information.

Threats

Threats include scams, malware, theft, accidental loss, service failure and unauthorized access.

Vulnerabilities

A vulnerability is a weakness, such as outdated software, an unlocked device or excessive permissions.

Controls

Controls reduce risk. Examples include updates, MFA, backups, access restrictions and awareness training.

Risk Thinking

A simple risk assessment considers likelihood and impact. High-impact risks deserve stronger attention even when they are not frequent.

Safe Guided Practical

1. Choose a fictional small training centre.
2. List five assets.
3. For each asset, identify one possible threat and one defensive control.
4. Rank the risks as Low, Medium or High using your reasoned judgement.

Review Questions

1. What is an asset?
2. Differentiate threat and vulnerability.
3. What is a security control?
4. What two factors are commonly considered in simple risk ranking?

Assignment

Create a risk register for five fictional assets.

LESSON 3: PASSWORDS, PASSPHRASES AND MFA

Account security begins with strong authentication and careful credential handling.

Account Security

Strong unique passwords + multi-factor authentication

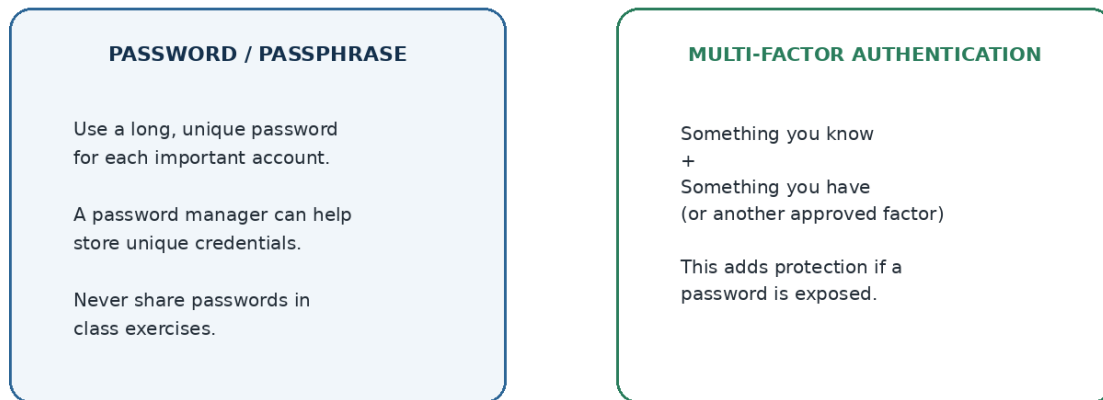


Figure 3. Passwords and multi-factor authentication.

Strong Password Practice

- Use long, unique passwords or passphrases.
- Do not reuse important passwords across services.
- Do not share passwords with classmates or colleagues.
- Avoid predictable personal information.

Password Managers

A reputable password manager can help generate and store unique passwords. Protect the manager itself with strong authentication.

Multi-Factor Authentication

MFA requires an additional approved factor beyond the password, reducing the impact of a stolen password.

Recovery Information

Keep recovery methods current and secure. Do not expose recovery codes in screenshots or class submissions.

Safe Guided Practical

1. Review the security settings of an account you are authorised to use.
2. Check whether MFA is available.
3. Create three fictional example passphrases; do not use real passwords.
4. Write a recovery checklist that excludes secret codes.

Review Questions

1. Why should passwords be unique?
2. What is MFA?
3. Why can password reuse be dangerous?
4. What information should never be submitted in a class exercise?

Assignment

Create an account-security policy for a fictional student portal.

LESSON 4: PHISHING AND SOCIAL ENGINEERING AWARENESS

Social engineering attempts to influence people into revealing information, sending money or taking unsafe actions.

Recognising Phishing

Check the sender, request, link context and urgency

From: account-update@example.invalid

Subject: URGENT - verify your account immediately

Your account will be closed today.

Click the link below and enter your password to keep access.

- 1 Unexpected urgency
- 2 Request for credentials
- 3 Suspicious or mismatched destination
- 4 Verify through an official channel instead

Figure 4. Warning signs in a fictional phishing message.

Common Warning Signs

- Unexpected urgency or threats
- Requests for passwords or one-time codes
- Unusual payment requests
- Mismatched sender identity
- Suspicious links or attachments
- Offers that seem unrealistic

Safe Verification

Do not rely only on the message itself. Verify important requests using an official website, known phone number or trusted person.

Phishing Beyond Email

Scams can arrive through SMS, messaging apps, social media, voice calls and fake websites.

Reporting

Use the organisation's reporting process. Reporting quickly can protect other users.

Safe Guided Practical

1. Review three fictional messages supplied by the instructor.
2. Highlight warning signs without opening any links.

3. Write the safe verification step for each message.
4. Draft a short phishing-report message.

Review Questions

1. What is phishing?
2. Name five warning signs.
3. Why should important requests be verified separately?
4. Where can phishing messages appear?

Assignment

Design a phishing-awareness poster.

LESSON 5: MALWARE AND ENDPOINT PROTECTION

Malware is software designed to cause harm, disrupt systems, steal information or enable unauthorized activity.

Malware Awareness

Know the categories; focus on prevention and recovery

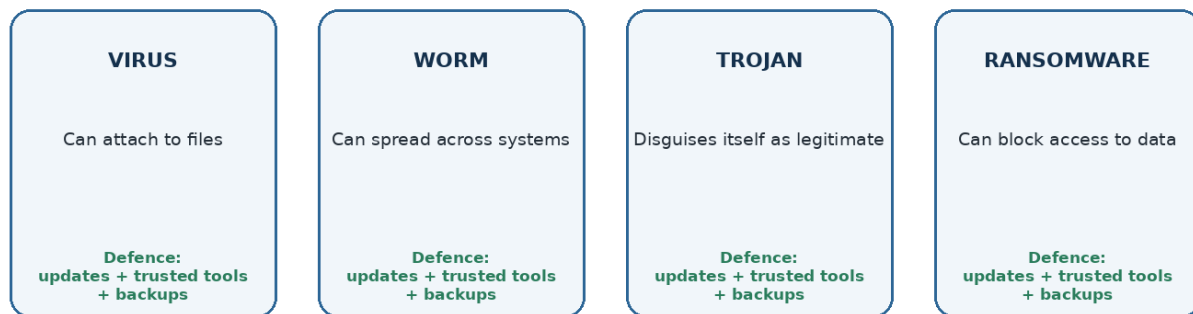


Figure 5. Malware categories and defensive measures.

Common Categories

- Virus
- Worm
- Trojan
- Ransomware
- Spyware

Endpoint Protection

Defensive measures include supported operating systems, timely updates, trusted security software, limited permissions and safe downloading habits.

Backups

Backups help recovery when files are lost, damaged or made unavailable.

Safe Learning Rule

Do not download, create or run malware for this course. Use diagrams and approved simulations only.

Safe Guided Practical

1. Classify five fictional malware scenarios by category.
2. Check whether your authorised device receives automatic security updates.
3. Identify where the operating system shows security status.

4. Write a prevention-and-recovery checklist.

Review Questions

1. What is malware?
2. What is ransomware?
3. Name four endpoint protections.
4. Why are backups important?

Assignment

Prepare a malware-awareness guide for beginners.

LESSON 6: UPDATES, PATCHING AND SECURE CONFIGURATION

Software updates can correct security weaknesses, fix defects and improve reliability.

Patch and Update Cycle

Updates close known weaknesses and improve reliability

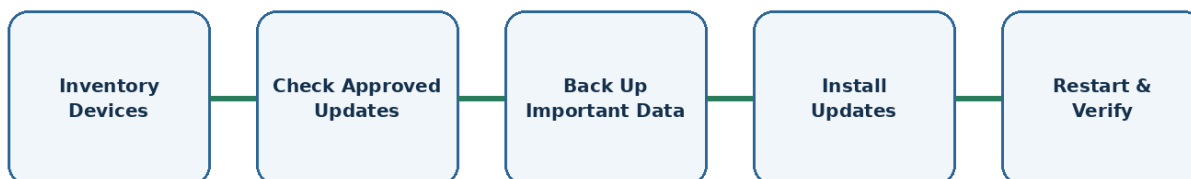


Figure 6. A safe patch and update cycle.

Patching

A patch is an update that corrects software problems, including security weaknesses.

Supported Software

Using supported software matters because unsupported products may stop receiving security fixes.

Secure Configuration

Examples include screen locks, automatic updates, approved applications and disabling unnecessary sharing features when policy requires it.

Change Carefully

Back up important work and follow school/work procedures before significant system changes.

Safe Guided Practical

1. Check the update status of an authorised device.
2. Record the operating-system version without exposing device identifiers.
3. Review screen-lock settings.
4. Create an update checklist for a fictional office.

Review Questions

1. What is a patch?

2. Why is supported software important?
3. Give three secure-configuration examples.
4. Why should important data be backed up before major changes?

Assignment

Create a monthly device-maintenance checklist.

LESSON 7: SAFE INTERNET, BROWSER AND NETWORK USE

Safe browsing and network habits reduce exposure to scams, unwanted downloads and information theft.

Safe Network Use

Think in layers: device, router/network, service and user



Figure 7. Layered safe network and internet use.

Browser Safety

- Keep the browser updated.
- Check the site address carefully.
- Use HTTPS as one signal of encrypted transport, while remembering it does not prove a site is trustworthy.
- Avoid unexpected downloads.

Public Wi-Fi

Public networks may not be appropriate for sensitive activity. Follow organisational policy and prefer trusted connections for important work.

Home or Office Network

Use approved router settings, strong administrative credentials and current firmware. Do not change network equipment you are not authorised to manage.

Downloads

Obtain software from trusted official sources and verify that it is appropriate for the device.

Safe Guided Practical

1. Compare two fictional website addresses and identify which one appears suspicious.
2. Check that your browser is current.
3. List activities you would avoid on an untrusted public network.

4. Create a safe-download checklist.

Review Questions

1. Does HTTPS alone prove a site is trustworthy?
2. Why are browser updates important?
3. What is a trusted network?
4. Why should software come from official sources?

Assignment

Write ten rules for safe internet use.

LESSON 8: DATA PROTECTION, PRIVACY AND CLASSIFICATION

Information should be handled according to its sensitivity, purpose and applicable rules.

Protecting Information

Classify information before deciding how to handle it

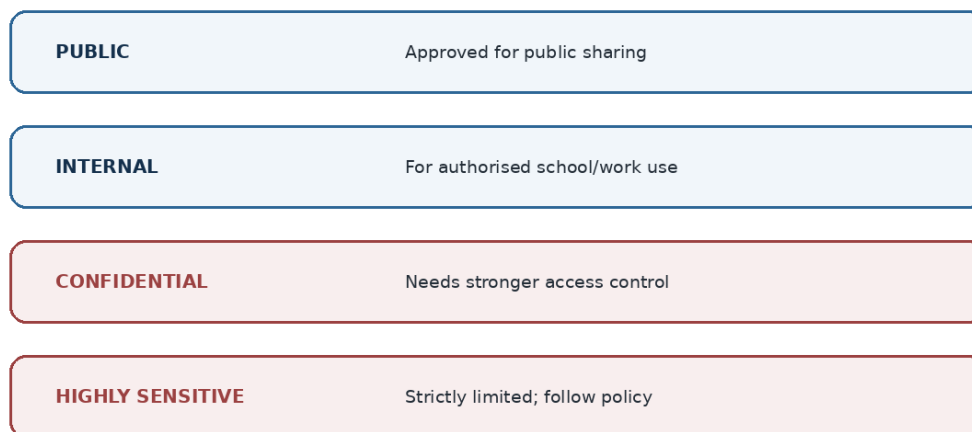


Figure 8. A simple information-classification model.

Personal Information

Examples can include names, contact details, identification numbers and account information. Collect only what is needed and protect it appropriately.

Classification

Organisations may label information Public, Internal, Confidential or another approved category.

Sharing

Before sharing, check the recipient, purpose, permissions and whether the chosen channel is appropriate.

Privacy

Privacy involves responsible collection, use, storage and sharing of information. Follow applicable policies and laws.

Safe Guided Practical

1. Classify ten fictional information items.
2. For each confidential item, state one appropriate protection.
3. Review a fictional sharing request and decide whether it should be approved, rejected or escalated.
4. Create a privacy checklist.

Review Questions

1. What is data classification?

2. Why should confidential data have stronger controls?
3. What should be checked before sharing information?
4. What does privacy involve?

Assignment

Create a simple data-handling guide for a training centre.

LESSON 9: BACKUPS AND RECOVERY

Backups provide independent copies of important information so work can be recovered after loss or disruption.

Backup Strategy

The 3-2-1 idea improves resilience



Backups should also be tested: a backup is useful only if it can be restored.

Figure 9. The 3-2-1 backup concept.

3-2-1 Concept

A common resilience idea is three copies of important data, using two types of storage, with one copy kept separately/offsite.

Backup Types

Organisations may use full, incremental or other backup methods depending on their needs.

Restore Testing

A backup should be tested because an unreadable or incomplete backup may not help during recovery.

Cloud Backup

Cloud services can support backup and synchronization, but access controls and account security still matter.

Safe Guided Practical

1. Identify three categories of files worth backing up.
2. Design a fictional 3-2-1 backup plan.
3. Create a restore-test checklist.
4. Explain how MFA helps protect cloud backups.

Review Questions

1. What is the purpose of a backup?

2. Explain the 3-2-1 concept.
3. Why should restores be tested?
4. Does cloud storage remove the need for account security?

Assignment

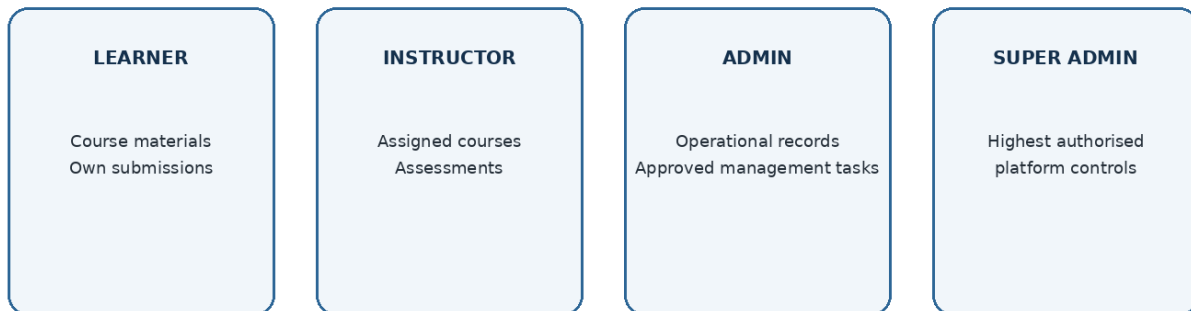
Design a backup plan for Ethan Digital Academy course records.

LESSON 10: ACCESS CONTROL AND LEAST PRIVILEGE

Access control determines who can use information or systems and what they are allowed to do.

Access Control

Give people only the access needed for their role



Principle of least privilege: provide only the permissions required.

Figure 11. Role-based access and least privilege.

Authentication and Authorization

Authentication verifies identity. Authorization determines permissions after identity is established.

Least Privilege

Users should receive only the access needed for their responsibilities.

Role-Based Access

Permissions can be assigned to roles such as learner, instructor and administrator.

Account Lifecycle

Access should be reviewed when people join, change roles or leave an organisation.

Safe Guided Practical

1. Create four fictional roles for a learning platform.
2. List what each role should and should not access.
3. Identify two examples of excessive permissions.
4. Create an access-review checklist.

Review Questions

1. Differentiate authentication and authorization.
2. What is least privilege?

3. What is role-based access?

4. When should access be reviewed?

Assignment

Design a role-permission matrix for a fictional LMS.

LESSON 11: EMAIL AND MESSAGING SECURITY

Email and messaging systems are important business tools and common channels for scams.

Recognising Phishing

Check the sender, request, link context and urgency

From: account-update@example.invalid

Subject: URGENT - verify your account immediately

Your account will be closed today.

Click the link below and enter your password to keep access.

- 1 Unexpected urgency
- 2 Request for credentials
- 3 Suspicious or mismatched destination
- 4 Verify through an official channel instead

Figure 4. Phishing awareness also applies to business email.

Attachment Safety

Unexpected attachments should be treated cautiously. Verify the sender and context before opening.

Business Email Requests

Payment, password-reset or account-change requests deserve independent verification.

Sensitive Information

Avoid sending secrets or unnecessary confidential data through unsuitable channels.

Account Compromise Signs

Unexpected sent messages, login alerts or unexplained account changes should be reported promptly.

Safe Guided Practical

1. Review four fictional email scenarios.
2. Decide which messages require verification.
3. Draft a safe response that does not reveal credentials.
4. Write an account-compromise reporting message.

Review Questions

1. Why are unexpected attachments risky?
2. Which requests deserve independent verification?

3. Name two possible signs of account compromise.
4. Should passwords be sent by ordinary email?

Assignment

Prepare a secure-email checklist.

LESSON 12: MOBILE DEVICE AND CLOUD SECURITY

Phones, tablets and cloud services hold valuable information and require the same security discipline as computers.

Safe Network Use

Think in layers: device, router/network, service and user



Figure 7. Security layers apply to devices, networks, services and users.

Mobile Security

- Use a screen lock.
- Keep the operating system and apps updated.
- Install apps from trusted sources.
- Review app permissions.
- Enable approved device-location or remote-lock features when appropriate.

Cloud Security

- Use strong authentication.
- Review sharing permissions.
- Remove public links that are no longer needed.
- Protect recovery methods.
- Follow organisation policy for confidential data.

Lost Devices

Report loss promptly and follow approved remote-lock, account-protection and recovery procedures.

Safe Guided Practical

1. Review app permissions on an authorised device.
2. Check whether automatic updates are enabled.
3. Review sharing settings on a non-sensitive cloud practice folder.

4. Create a lost-device response checklist.

Review Questions

1. Name four mobile-security practices.
2. Why should app permissions be reviewed?
3. What should be checked on cloud-sharing links?
4. What should happen after a device is lost?

Assignment

Create a mobile-and-cloud security guide.

LESSON 13: SECURITY AWARENESS FOR ORGANISATIONS

Technical controls are stronger when people understand their responsibilities and know how to report concerns.

Recognising Phishing

Check the sender, request, link context and urgency

From: account-update@example.invalid

Subject: URGENT - verify your account immediately

Your account will be closed today.

Click the link below and enter your password to keep access.

- 1 Unexpected urgency
- 2 Request for credentials
- 3 Suspicious or mismatched destination
- 4 **Verify through an official channel instead**

Figure 4. Recognising and reporting suspicious communication is a core awareness skill.

Awareness Topics

- Phishing
- Password and MFA practices
- Data handling
- Safe remote work
- Incident reporting
- Physical security

Training Culture

Security training should be practical, understandable and repeated. Staff should be encouraged to report mistakes quickly.

Clear Procedures

Users need simple instructions for common situations such as suspicious email, lost devices and unusual login alerts.

Measurement

Organisations can measure completion of training, reporting rates and improvement in safe behaviours without creating a culture of blame.

Safe Guided Practical

1. Design a one-month awareness plan.

2. Choose four weekly topics.
3. Create one short safety message for each week.
4. Define two positive measures of programme effectiveness.

Review Questions

1. Why is awareness training important?
2. Name five awareness topics.
3. Why should reporting be encouraged?
4. How can awareness programmes be measured?

Assignment

Create a cybersecurity-awareness calendar.

LESSON 14: INCIDENT RECOGNITION, REPORTING AND RESPONSE

A security incident is an event that may threaten confidentiality, integrity or availability and requires an organised response.

Basic Incident Response

A safe beginner workflow for suspected security incidents



Figure 10. A safe beginner incident-response workflow.

Possible Warning Signs

- Unexpected account changes
- Unusual login alerts
- Missing or inaccessible files
- Suspicious messages sent from an account
- Lost device
- Unexpected security warnings

Immediate Actions

Follow the organisation's approved procedure. Limit further exposure when safe, report promptly and avoid destroying useful evidence.

Escalation

Learners and ordinary users should contact the responsible administrator, teacher, parent/guardian or support person rather than attempting risky technical intervention.

Recovery

Recovery can include password resets through official channels, restoring backups, updating systems and reviewing what happened.

Safe Guided Practical

1. Read a fictional lost-laptop scenario.

2. Write the first five safe actions.
3. Read a fictional compromised-email scenario.
4. Draft an incident report containing time, observed symptoms and actions already taken.

Review Questions

1. What is a security incident?
2. Name four warning signs.
3. Why should evidence not be casually deleted?
4. Who should a learner contact during a suspected incident?

Assignment

Create an incident-report form for a training centre.

LESSON 15: CYBERSECURITY POLICIES, ETHICS AND LAW

Cybersecurity professionals must work within authorization, policy, ethics and applicable law.

Threat, Vulnerability and Risk

A simple defensive risk model



Figure 2. Security risk management begins with clear assets, responsibilities and controls.

Authorization

Only access systems, accounts and information you are explicitly authorised to use.

Responsible Practice

Do not test security controls on real people or systems without proper written permission and an approved scope.

Policies

Organisations use acceptable-use, password, data-handling, backup, incident-response and access-control policies.

Professional Ethics

Protect confidentiality, avoid conflicts of interest, document work accurately and escalate concerns responsibly.

Legal Awareness

Privacy, computer-misuse and data-protection requirements vary by jurisdiction. Professionals must follow applicable law and organisational guidance.

Safe Guided Practical

1. Review a fictional acceptable-use policy.
2. Identify five behaviours that require authorization.

3. Rewrite three vague rules into clear security-policy statements.
4. Discuss why ethical behaviour matters even when a technical action is possible.

Review Questions

1. What is authorization?
2. Why is written scope important for security testing?
3. Name four cybersecurity policies.
4. What does professional ethics require?

Assignment

Draft a one-page acceptable-use policy.

LESSON 16: RISK ASSESSMENT AND BUSINESS CONTINUITY

Organisations prepare for disruption by identifying important services, evaluating risks and planning how to continue or recover operations.

Threat, Vulnerability and Risk

A simple defensive risk model



Figure 2. Risk assessment connects assets, threats, weaknesses and controls.

Risk Register

A risk register records assets, threats, weaknesses, existing controls, likelihood, impact and planned actions.

Business Impact

Some systems are more critical than others. Prioritise services that are essential to safety, learning, payments or operations.

Continuity

Business continuity focuses on maintaining essential functions during disruption.

Disaster Recovery

Disaster recovery focuses on restoring technology and data after a serious event.

Review

Plans should be tested and updated as systems and responsibilities change.

Safe Guided Practical

1. Create a five-item risk register for a fictional academy.
2. Identify the two most important services.
3. Write a simple continuity plan for an internet outage.
4. Write a simple recovery checklist for accidental file loss.

Review Questions

1. What is a risk register?
2. Differentiate business continuity and disaster recovery.
3. Why should critical services be prioritised?
4. Why must plans be tested?

Assignment

Prepare a basic business-continuity plan.

LESSON 17: CYBERSECURITY CAREERS AND DEFENSIVE TOOLS

Cybersecurity includes technical and non-technical professional pathways.

Cybersecurity Career Pathways

Defensive learning can lead to many professional roles



Figure 12. Examples of defensive cybersecurity career pathways.

Career Areas

- Security awareness and training
- Security Operations / Blue Team
- Governance, Risk and Compliance (GRC)
- Cloud security
- Identity and access management
- Digital forensics under proper authorization
- Security engineering

Defensive Tools

Professionals may use endpoint protection, logging, vulnerability-management platforms, identity systems, backup tools and monitoring dashboards. Tool use must follow authorization and policy.

Skills

Communication, documentation, risk thinking, operating-system knowledge, networking fundamentals and ethical judgement are important.

Learning Path

Build strong computer, networking and operating-system foundations before moving into advanced security specialisations.

Safe Guided Practical

1. Choose three cybersecurity careers.

2. Research the responsibilities using instructor-approved resources.
3. List the foundational skills for each.
4. Create a six-month defensive-learning plan.

Review Questions

1. Name four cybersecurity careers.
2. What does a blue team focus on?
3. What is GRC?
4. Why does tool use require authorization?

Assignment

Prepare a career pathway presentation.

LESSON 18: FINAL DEFENSIVE SECURITY PROJECT

The final project applies the course to a fictional organisation without attacking any real system.

Cybersecurity Foundations

Confidentiality, Integrity and Availability

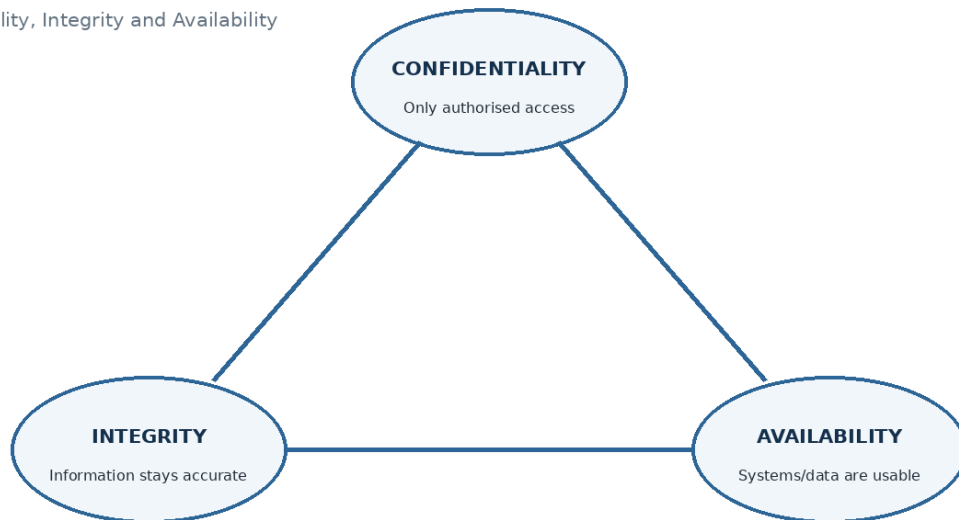


Figure 1. Cybersecurity balances confidentiality, integrity and availability.

Scenario

Ethan Digital Academy wants a beginner-friendly cybersecurity improvement plan for its learning environment.

Required Sections

- Asset list
- Simple risk register
- Account-security standard
- Phishing-awareness guide
- Device-update checklist
- Data-classification guide
- Backup plan
- Role-based access matrix
- Incident-reporting procedure
- Security-awareness calendar

Presentation

Summarise the plan in a professional report and short presentation.

Safety Boundary

Do not scan, probe, exploit, bypass or test live systems. The project is documentation, defensive review and safe configuration awareness only.

Safe Guided Practical

1. Create the ten required project sections.

2. Use fictional or non-sensitive example data.
3. Add diagrams and checklists.
4. Review the plan for clarity and practicality.
5. Present the plan as if briefing academy management.

Review Questions

1. What is the purpose of the final project?
2. Why does it avoid live-system testing?
3. Which parts address prevention?
4. Which parts address recovery?

Assignment

Submit the complete Cybersecurity Improvement Plan.

REAL-LIFE CASE STUDY - ETHAN DIGITAL ACADEMY

A digital academy stores learner records, course materials, payment information and staff accounts. Management introduces role-based access, MFA for privileged accounts, regular updates, staff awareness training, controlled data sharing, tested backups and a clear incident-reporting process. The objective is not to make risk disappear completely, but to reduce preventable incidents and improve recovery when problems occur.

FINAL PRACTICAL PROJECT - 100 MARKS

1. Prepare an asset inventory for the fictional academy.
2. Create a five-risk register with defensive controls.
3. Write an account-security standard.
4. Create a phishing-awareness guide.
5. Create an update and device-security checklist.
6. Create a data-classification and sharing guide.
7. Design a 3-2-1 backup plan.
8. Create a role-based access matrix.
9. Create an incident-report form and response flow.
10. Present the complete Cybersecurity Improvement Plan.

QUIZ - 30 QUESTIONS

1. What is cybersecurity?
2. What does confidentiality mean?
3. What does integrity mean?
4. What does availability mean?
5. What is an asset?
6. What is a threat?
7. What is a vulnerability?
8. What is a security control?
9. Why should passwords be unique?
10. What is MFA?
11. What is phishing?
12. Name three phishing warning signs.

13. What is malware?
14. What is ransomware?
15. Why are software updates important?
16. What is a patch?
17. Does HTTPS alone prove a site is trustworthy?
18. What is data classification?
19. What is the purpose of a backup?
20. Explain the 3-2-1 backup concept.
21. Differentiate authentication and authorization.
22. What is least privilege?
23. Why should cloud-sharing permissions be reviewed?
24. Name three possible signs of a security incident.
25. Why should incidents be reported promptly?
26. What is authorization in cybersecurity?
27. What is a risk register?
28. Differentiate business continuity and disaster recovery.
29. Name three cybersecurity career areas.
30. Why must defensive tools be used only with authorization?

FINAL EXAMINATION - 100 MARKS

Section A - Objective Questions (20 Marks)

Twenty questions covering cybersecurity terminology, account security, phishing, malware, data protection, backups, access control and incident response.

Section B - Short Answer and Case Analysis (30 Marks)

Explain concepts and recommend safe defensive actions for realistic school and business scenarios.

Section C - Defensive Practical Assessment (50 Marks)

Prepare selected parts of a cybersecurity improvement plan using fictional information: risk register, account-security checklist, data classification, backup plan, access matrix and incident-reporting procedure.

COURSE COMPLETION OUTCOME

Graduates of EDA-CYB-114 Cybersecurity Fundamentals will understand essential defensive cybersecurity principles and apply safer digital habits in school, office and business environments. They will be prepared for deeper study in networking, security operations, governance and risk, cloud security and other authorised defensive specialisations.